

Earth, Environment and Society
ENVIRSC 1C03
Climate, Water and Environment
Winter 2027



SCIENCE

Instructor Information

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Office Hours:

TBD

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Course Information

Lectures: Tuesday, Wednesday, Friday 3:30 - 4:20

Labs: 2 hour lab per week.

This course is an introduction to environmental science and processes, including topics as Earth's global energy balance, climate change, the hydrological cycle, soils management, water pollution and eutrophication. It also introduces the principles of sustainability and how science can be used to better understand the impacts of humanity's interaction with the environment.

Specifically, the course introduces the functions of Earth's atmosphere and its composition, and Earth's global energy balance; the quantitative approaches to understand the hydrologic cycle, and their use in water resource management; it presents an overview of the global atmospheric and oceanic circulation and the delineation of climatic zones; soil-water processes, the major elemental cycles, ecosystem functions and the role of biodiversity.

The course highlights the impact of human activity on life-support systems through discussions of the anthropogenic greenhouse effect and its link to climate change, atmospheric and water pollution, urban heat islands, soil degradation, biodiversity loss, the health effects of pollutants, the management of the footprint cities, and humanity's ecological footprint.

The course allows students through hands-on field-based activities and problem-solving exercises, to interpret Earth's radiative budget, water balances, biogeographical differences, carbon sequestration potential, soil properties, and box models, and apply them to real-world problems. Finally, the exploration of new modes of communication and the analysis of scientific information through an inquiry-based assignment are features of this course.

Course Dates: 01/04/2027 - 04/07/2027

Units: 3.00

Course Delivery Mode: In Person

Course Description: An introduction to the science of environmental issues and sustainability through the study of the soil, climate and water processes. Lectures, web modules (three hours), one lab (two hours); one term This course may be offered in-person or online. Students are encouraged to check scheduling information and the course outline to confirm the course offering details.

Important Links

- [Mosaic](#)
- [Avenue to Learn](#)
- [Student Accessibility Services - Accommodations](#)
- [McMaster University Library](#)
- [eReserves](#)
- [Open Education Resources](#)

Required Materials and Texts

Please sign in with your MacID [here](#) to view your booklist

There is no textbook to purchase. We will use an Open Education Resource textbook. Further details are posted on avenue.

Please note that a subscription fee must be paid to register the iClicker app, which will be used to record in-class participation.

Class Format

In Person

Course Evaluation

Course Evaluation

Assessment	Weight
Labs (5 @ 5%)	25%
Quizzes, best 4 out of 5 @4% each	16%
In-class participation (iClicker)	10%
Infographic Term Project - individual	15%
Final Exam	34%

Undergraduate Grading Scale

The grades breakdown for McMaster's 12-Point Grading Scale

Grade	Equivalent Grade Point	Equivalent Percentages
A+	12	90-100
A	11	85-89
A-	10	80-84
B+	9	77-79
B	8	73-76
B-	7	70-72
C+	6	67-69
C	5	63-66
C-	4	60-62
D+	3	57-59
D	2	53-56
D-	1	50-52
F	0	0-49

Course Schedule

The **lecture schedule** for this course is posted on avenue and outlines what topics will be covered and when they will be covered during the term. Please note that the

lecture schedule will likely be modified throughout the term as some topics may not be finished in the predicted lecture time

Laboratory Overview, Operation, and Safety

Labs begin the **week of January 18**. The time of your lab will depend on which lab section you are enrolled into on Mosaic. Please see your Mosaic timetable for this information.

Turnitin.com

Some courses may use a web-based service (Turnitin.com) to reveal authenticity and ownership of student submitted work. For courses using such software, students will be expected to submit their work electronically either directly to Turnitin.com or via an online learning platform (e.g. A2L, etc.) using plagiarism detection (a service supported by Turnitin.com) so it can be checked for academic dishonesty.

Students who do not wish their work to be submitted through the plagiarism detection software must inform the Instructor before the assignment is due. No penalty will be assigned to a student who does not submit work to the plagiarism detection software. All submitted work is subject to normal verification that standards of academic integrity have been upheld (e.g., on-line search, other software, etc.). For more details about McMaster's use of Turnitin.com please go to www.mcmaster.ca/academicintegrity.

Generative AI: Use Prohibited

Students are not permitted to use generative AI in this course. In alignment with [McMaster academic integrity policy](#), it "shall be an offence knowingly to ... submit academic work for assessment that was purchased or acquired from another source". This includes work created by generative AI tools. Also stated in the policy is the following, "Contract Cheating is the act of "outsourcing of student work to third parties" (Lancaster & Clarke, 2016, p. 639) with or without payment." Using Generative AI tools is a form of contract cheating. Charges of academic dishonesty will be brought forward to the Office of Academic Integrity.

Checking Grades & Regrade Policy

Checking Your Grades

Your marks will be recorded on Avenue. It is your responsibility to check that all grades entered into Avenue are recorded properly. All grade concerns and discrepancies must be reported to the Instructional Assistant within a week of receiving the grade.

APPROVED ADVISORY STATEMENTS

Academic Integrity

You are expected to exhibit honesty and use ethical behaviour in all aspects of the learning process. Academic credentials you earn are rooted in principles of honesty and academic integrity. **It is your responsibility to understand what constitutes academic dishonesty.**

Academic dishonesty is to knowingly act or fail to act in a way that results or could result in unearned academic credit or advantage. This behaviour can result in serious consequences, e.g. the grade of zero on an assignment, loss of credit with a notation on the transcript (notation reads: "Grade of F assigned for academic dishonesty"), and/or suspension or expulsion from the university. For information on the various types of academic dishonesty please refer to the [Academic Integrity Policy](#).

The following illustrates only three forms of academic dishonesty:

- plagiarism, e.g. the submission of work that is not one's own or for which other credit has been obtained.
- improper collaboration in group work.
- copying or using unauthorized aids in tests and examinations.

Authenticity / Plagiarism

Some courses may use a web-based service (Turnitin.com) to reveal authenticity and ownership of student submitted work. For courses using such software, students will be expected to submit their work electronically either directly to Turnitin.com or via an online learning platform (e.g. Avenue to Learn, etc.) using plagiarism detection (a service supported by Turnitin.com) so it can be checked for academic dishonesty.

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Courses with an On-line Element

Some courses may use on-line elements (e.g. e-mail, Avenue to Learn, LearnLink, web pages, capa, Moodle, ThinkingCap, etc.). Students should be aware that, when they access the electronic components of a course using these elements, private information such as first and last names, user names for the McMaster e-mail accounts, and program affiliation may become apparent to all other students in the same course. The available information is dependent on the technology used. Continuation in a course that uses on-line elements will be deemed consent to this disclosure. If you have any questions or concerns about such disclosure please discuss this with the course instructor.

Online Proctoring

Some courses may use online proctoring software for tests and exams. This software may require students to turn on their video camera, present identification, monitor and record their computer activities, and/or lock/restrict their browser or

other applications/software during tests or exams. This software may be required to be installed before the test/exam begins.

Conduct Expectations

As a McMaster student, you have the right to experience, and the responsibility to demonstrate, respectful and dignified interactions within all of our living, learning and working communities. These expectations are described in the [Code of Student Rights & Responsibilities](#) (the “Code”). All students share the responsibility of maintaining a positive environment for the academic and personal growth of all McMaster community members, **whether in person or online**.

It is essential that students be mindful of their interactions online, as the Code remains in effect in virtual learning environments. The Code applies to any interactions that adversely affect, disrupt, or interfere with reasonable participation in University activities. Student disruptions or behaviours that interfere with university functions on online platforms (e.g. use of Avenue 2 Learn, WebEx or Zoom for delivery), will be taken very seriously and will be investigated. Outcomes may include restriction or removal of the involved students’ access to these platforms.

Academic Accommodation of Students with Disabilities

Students with disabilities who require academic accommodation must contact [Student Accessibility Services](#) (SAS) at 905-525-9140 ext. 28652 or sas@mcmaster.ca to make arrangements with a Program Coordinator. For further information, consult McMaster University’s [Academic Accommodation of Students with Disabilities](#) policy.

Requests for Relief for Missed Academic Term Work

In the event of an absence for medical or other reasons, students should review and follow the [Policy on Requests for Relief for Missed Academic Term Work](#).

Academic Accommodation for Religious, Indigenous, or Spiritual Observances (RISO)

Students requiring academic accommodation based on religious, Indigenous or spiritual observances should follow the procedures set out in the [RISO](#) policy. Students should submit their request to their Faculty Office ***normally within 10 working days*** of the beginning of term in which they anticipate a need for accommodation or to the Registrar's Office prior to their examinations. Students should also contact their instructors as soon as possible to make alternative arrangements for classes, assignments, and tests.

Copyright and Recording

Students are advised that lectures, demonstrations, performances, and any other course material provided by an instructor include copyright protected works. The Copyright Act and copyright law protect every original literary, dramatic, musical and artistic work, **including lectures** by University instructors.

The recording of lectures, tutorials, or other methods of instruction may occur during a course. Recording may be done by either the instructor for the purpose of authorized distribution, or by a student for the purpose of personal study. Students should be aware that their voice and/or image may be recorded by others during the class. Please speak with the instructor if this is a concern for you.

Extreme Circumstances

The University reserves the right to change the dates and deadlines for any or all courses in extreme circumstances (e.g., severe weather, labour disruptions, etc.). Changes will be communicated through regular McMaster communication channels, such as McMaster Daily News, Avenue to Learn and/or McMaster email.